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Palaeomagnetic results from the late Precambrian Chela Group of southwest Angola

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ABSTRACT

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Palaeomagnetic results from fourteen sites (56 samples) in the late Precambrian Chela Group of southwest Angola are reported. After alternating field demagnetization eight of the sites have mean directions closely grouped around $D=123°$, $I=-44\frac{1}{2}°$; the dispersion of these directions increases substantially after tectonic corrections suggesting magnetic overprinting.

Intensive thermal demagnetization proved more effective than alternating field demagnetization in resolving magnetic components. After removal of viscous components, samples from five sites (Group I) show a single component, similar to that observed after alternating field demagnetization. At four more sites (Group II) this same component is present together with another higher-temperature component. The mean direction for the component observed at all nine sites is $D=119.7°$, $I=-45.9°$ with $k=40.5$. Tectonic corrections produced little change in mean direction but reduced k to 4.5; the data supports the magnetic overprint hypothesis. A high blocking temperature component was resolved by remagnetization circle analysis of three flat-lying sites (Group III) and this component is similar to the higher-temperature component observed at site DG (Group II) *after* tectonic correction has been made. The component observed in Group III samples ($D=311°$, $I=49°$), confirmed by some samples from Group II, is similar to that observed in the Nosib Group just to the south in Namibia.

It is suggested that the mean Group III result gives the primary direction of the Chela Group, that the data support the correlation of the Nosib and Chela Groups and that the Chela rocks were overprinted magnetically during an event subsequent to their tilting.

Introduction

The Chela Group is a volcano-sedimentary platform assemblage preserved in isolated areas around and south of Lubango (formerly Sa da Bandeira) in southwest Angola (Fig. 1). It was deposited in late Precambrian times on granitoids, schists and gneisses of the southern Angola basement, part of the Congo Craton.

It was Correia (1973), working on the Humpata Plateau just south of Lubango, who revealed the presence of widespread volcaniclastic beds, while showing the basal Chela conglomerates were not of fluvio-glacial origin as previously thought (Vale, 1971, 1973a, b). This led to a complete redefinition of the Chela Group stratigraphy and nomenclature (Correia, 1976; Kröner and Correia, 1980). The

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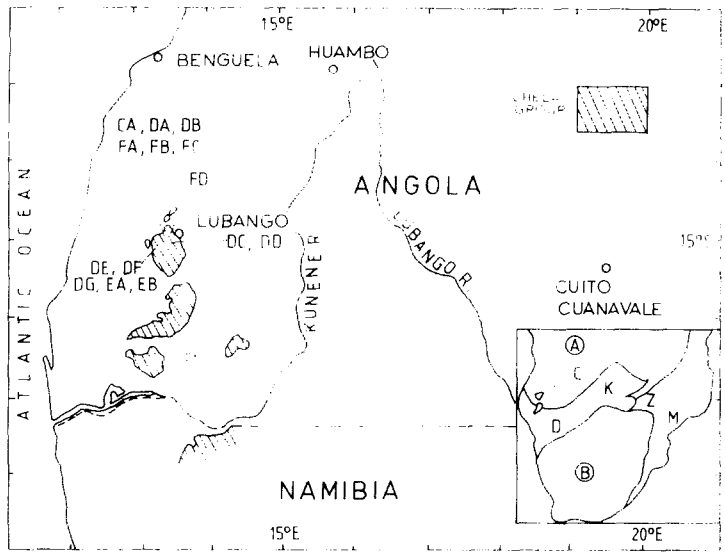


Fig. 1. Sketch map of southern Angola showing Chela Group occurrences and palaeomagnetic sampling sites. The inset shows the Congo (*A*) and Kalahari (*B*) cratons of southern Africa. The locations of the Chela Group rocks (*C*) and some Pan-African mobile belts are indicated: *D*=Damara Belt; *K*=Katanga Belt; *Z*=Zambezi Belt; *M*=Mozambique Belt.

basal Tundavala Formation consists almost exclusively of arenaceous rocks, whereas the overlying Humpata Formation is characterised by thick, very fine grained, volcaniclastic strata of rhyolitic composition which display vitroclastic features. The Humpata rocks were deposited in a sub-aqueous environment. Intercalations of two persistent beds of purplish-red, haematitic and partly welded rhyolite tuff at the base and the top of this formation serve as excellent stratigraphic markers. The overlying Bruco Formation is again mainly arenaceous in composition, and the Cangalongue Formation contains shales, siltstones, limestones and ferruginous sandstones. Separated from the Chela Group as the Leba Formation is a homogeneous carbonate sequence with stromatolitic dolostones and flaggy chert (Correia, 1976), which overlies the other strata paraconformably and appears to be part of a distinct and younger tectono-sedimentary cycle. The Chela and Leba rocks were not obviously involved in the events of the nearby Pan-African (900–550 Ma) Damara orogeny which involved rocks just to the south and they are neither penetratively deformed nor metamorphosed. Monoclinal folding is localised along the main E–W, NW–SE and N–S fault directions and minor small-scale folds are said to be due to synsedimentary slumping.

Beetz (1933) considered the Chela Group sediments to be early Palaeozoic and correlated them with the Otavi and Nama Groups of Namibia. De Carvalho (1972) proposed a correlation of the Chela–Leba succession with the Nosib and Otavi Groups of the miogeoclinical Damara sequence in northern Namibia, while Kröner (1974) and Kröner and Correia (1980), following detailed lithostratigraphic work there and in southern Angola, were more precise and equated the Chela Group with the Nosib Group and the Leba Formation with the middle Tsumeb Maieberg and Elandshoek Formations of the Otavi Group.

The age of the Chela and Leba sequences is still uncertain. The Chela Group rests unconformably upon granitoid basement for which De Carvalho (1972) quotes Rb–Sr biotite ages of 1023 to 1040±60 Ma while Torquato and Amaral (1973) report whole-rock and biotite K–Ar ages between 1037±38 and 1085±43

Ma for basement granites and diorites. The latter also report a K–Ar whole-rock age of 777 ± 41 Ma for a noritic dyke cutting Chela sediments and 800 ± 13 Ma from a granodiorite associated with the considerable number of dykes that intrude the Chela–Leba rocks of the Humpata Plateau. These (inadequate) data suggest the Chela–Leba sequence is probably older than about 780 Ma and is possibly younger than about 1060 Ma.

We report here palaeomagnetic measurements made on Chela sediments from the Humpata Plateau. The original purpose of the study was to test the correlation of the Chela Group with the Nosib Group (where complementary palaeomagnetic work was performed by McWilliams and Kröner (1981) who also give an account of the tectonic development of the region) and to obtain data that would help to define more clearly the apparent polar wander path (APWP) for Africa (or, at least, the Congo Craton) in the interval 1100–800 Ma.

Sampling

Cored samples (56), 25 mm in diameter, all oriented by sun compass, were obtained by A. Kröner from fourteen sites (Fig. 1) spread throughout the Chela Group (Table 1). All samples were reddish in some degree suggesting haematite pigment and were predominantly from pyroclastic beds (ten sites) with two sites each in quartzites (CA and FB) and thinly bedded shales (EA and EB).

Measurements

Two, and often three, cylindrical specimens (25×25 mm) were cut from each sample core. All alternating magnetic field demagnetization (AFD) experiments were performed in Harare using two-axis tumbling equipment (McElhinny, 1966) with remanence measurements made using either a commercial spinner magnetometer (PAR SM1) or the spinner described by Van Oorschot and Ridler (1976).

TABLE 1

Sampling details of the Chela Group

Site	Lat. (S)	Long. (E)	N	Bedding	Formation
CA	15°3′	13°32′	5	68°/281°	Tundavala
DA	14°50′	13°24′	4	3°/79°	Humpata
DB	14°50′	13°24′	5	4°/321°	Humpata
DC	14°50′	13°30′	4	8°/159°	Humpata
DD	14°56′	13°28′	4	12°/240°	Humpata
DE	15°4′	13°14′	4	56°/150°	Humpata
DF	15°4′	13°14′	4	0/0	Humpata
DG	15°4′	13°14′	2	61°/138°	Humpata
EA	15°6′	13°18′	5	3°/202°	Cangalongue
EB	15°6′	13°18′	3	5°/232°	Cangalongue
FA	14°49′	13°23′	4	0/0	Bruco
FB	14°49′	13°23′	4	0/0	Bruco
FC	14°50′	13°23′	4	0/0	Bruco
FD	14°54′	13°16′	4	0/0	Bruco

Notes: N is the number of cores drilled. Column 5 gives the dip and direction of dip of beds.

Early thermal demagnetization (TD) experiments were carried out in Harare and Canberra using apparatus in which field cancellation within the furnace is produced by three orthogonal coil systems involving ten square Rubens coils and monitored by three fluxgate detectors. Extensive and detailed TD experiments were carried out at Stanford University in 1986 using a commercially built furnace (Schonstedt TSD-1) in which field cancellation is achieved by use of mumetal shields; remanence measurements were made with an ScT cryogenic magnetometer. In all three laboratories the cooling of heated specimens was carried out in air, and field cancellation to within ± 5 gammas (nT) of zero is claimed. Because the data obtained at Stanford are considerably less noisy than those from the other laboratories (where it is thought noise originated in the furnaces), the results from the Stanford experiments are reported here.

Results

Natural remanent magnetization (NRM)

Magnetic intensities, as would be expected for the different lithologies involved, varied

greatly and ranged over four orders of magnitude. Intensities of specimens from the weakly magnetised site DF were typically 2×10^{-4} A/m while those for FD were around 0.5 A/m.

Directions of magnetization in different specimens from the same core were generally very similar. The NRM site mean directions, without tectonic corrections, for those specimens that subsequently provided the results quoted here from AFD and TD experiments are given, respectively, in Tables 2 and 3. (The NRM and AFD results in Table 2 are based upon one specimen per sample. For Table 3, where applicable, the results from two specimens from the same sample were combined before site means were calculated). Figure 2a shows that, with the exceptions of sites DC and FA to FD inclusive, NRM site mean directions have negative inclinations and are confined essentially to the southeast quadrant of the stereonet.

Alternating field demagnetization

The reddish colouration of samples suggested haematite could be a carrier of remanence, a suggestion supported by AFD of pilot specimens which showed little or no change in intensity. As a result, AFD was limited in favour of extensive TD which it was felt would be more effective in resolving the components of remanence.

Generally, two specimens from each site were demagnetized progressively to 50 mT. In the majority of cases, after removal of a low-coercivity component, the pilot specimens showed little or no changes in direction *or* intensity. The viscous component was random from specimen to specimen and only in about one third of the specimens could it be related to the Earth's present field in southern Angola [declination (D) = 346°; inclination (I) = −54°]. The magnitude of the change in direction was small except in the case of the

TABLE 2

NRM and AFD site mean directions (without tectonic corrections) for the Chela Group

Site	N	NRM				AFD					
		D (deg)	I (deg)	R	k	B (mT)	D (deg)	I (deg)	R	k	α_{95} (deg)
CA	5	104.1	−50.7	4.921	50.4	20	99.4	−45.2	4.956	91.9	8.0
DA	4	148.7	−45.0	3.979	145	30	135.9	−42.7	3.971	102	9.1
DB	5	158.4	−52.0	4.973	147	20	141.0	−58.6	4.990	399	3.8
DC	4	184.4	+ 6.2	3.224	3.9	30	144.8	−30.9	3.862	21.8	20.1
DD	4	114.8	−37.9	3.993	446	20	116.2	−43.1	3.995	564	3.9
DE	4	105.0	−45.4	3.945	54.3	30	99.2	−56.6	3.914	35.1	15.7
DF	4	164.3	−28.4	3.413	5.1						
DG	2	88.4	−58.6	1.964	27.4						
EA	5	115.0	−45.4	4.953	85.2	30	118.2	−40.2	4.951	82.2	8.5
EB	3	117.6	−31.8	2.796	9.8	30	123.3	−30.6	2.858	14.1	34.1
FA	4	12.0	+22.5	3.970	98.2	30	11.1	+23.4	3.965	85.3	10.0
FB	4	211.1	−53.7	3.719	10.7						
FC	4	169.3	+ 8.6	1.859[a]							
FD	4	204.1	−48.4	3.858	21.1	20	206.5	−49.1	3.869	22.9	19.6

Notes: D and I are the declination and inclination of the resultant vector of the N unit (specimen = sample) remanence vectors: R is its magnitude, k is the estimate of precision (Fisher, 1953) while a_{95} is the semi-angle of the cone of 95% confidence for the mean direction.

[a]The "resultant" of data which are random on Watson's (1956) criterion.

TABLE 3

NRM and TD site mean directions (without tectonic corrections) for the Chela Group

Group	Site	N	n	NRM				Range (°C)	TD				
				D (deg)	I (deg)	R	k		D (deg)	I (deg)	R	k	α_{95}
I	CA	5	5	104.1	−50.7	4.921	50.6	20/405 to origin	109.3	−42.2	4.924	52.6	10.6
I	DA	4	7	150.2	−45.3	3.990	300	510/615 to origin	134.2	−51.9	3.987	231	6.1
I	DB	5	9	159.4	−51.6	4.962	105	200/590 to origin	138.0	−60.2	4.994	667	3.0
I	DC	4	7	177.1	− 9.3	3.845	19.4	200/640 to origin	137.0	−45.2	3.941	50.8	13.0
I	DD	4	5	121.1	−39.3	3.990	300	350/600 to origin	121.1	−44.2	3.991	333	5.0
II	DE	4	4	105.0	−45.4	3.945	54.6	20/210 to 570/660	101.8	−46.1	3.920	37.5	15.2
I	DF	3	4	164.4	−44.1	2.630	5.4		163.9	−44.3	2.943	35.1	21.1
II	DG	2	4	87.6	−59.8	1.973	37.0	20/310 to 620/670	102.8	−45.1	1.925	13.3	74.9
II	EA	5	7	116.3	−46.5	4.955	88.9	20/500 to 670/origin	112.7	−41.8	4.954	87.0	8.3
II	EB	3	3	117.6	−31.8	2.796	9.8	20/410 to 665/origin	126.8	−29.5	2.895	19.0	29.1
I	FD	4	4	204.4	−58.4	3.920	37.5	20/450 to origin	206.9	−56.4	3.911	33.7	16.1

Notes: see Table 2; n is number of specimens.

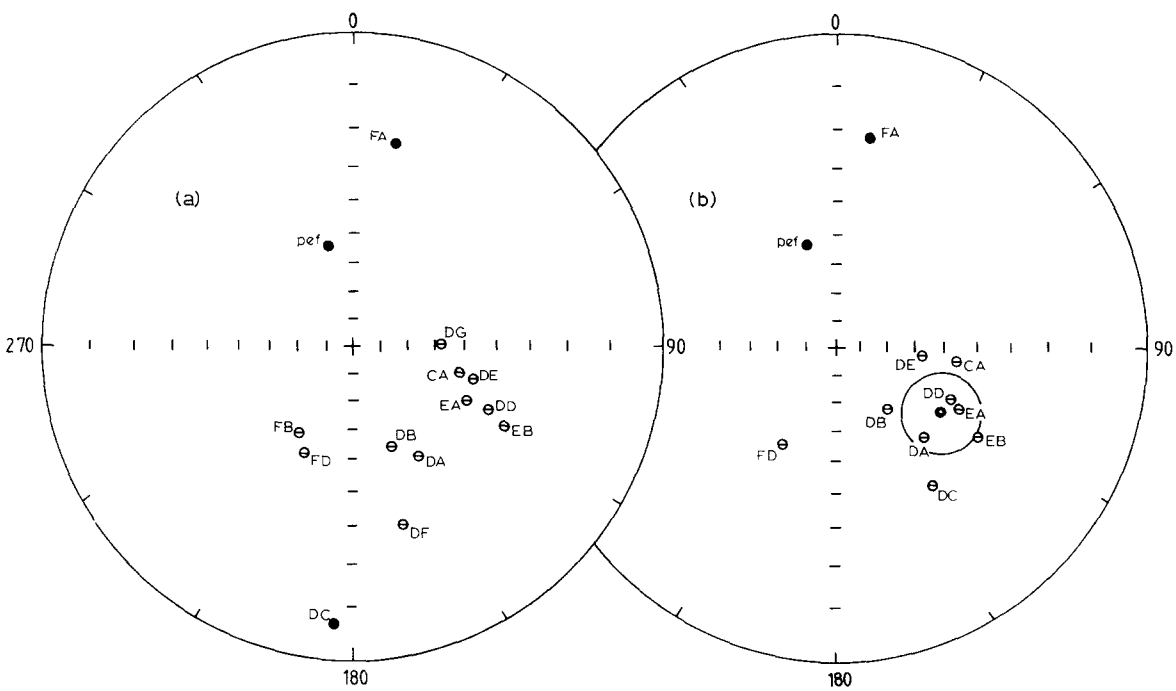


Fig. 2. Stereographic projections showing site mean directions (a) at NRM and (b) after limited AFD. Circles denote negative inclinations; dots, positive inclinations; *pef* shows the direction of the present earth's field. The mean direction (with circle of confidence) of the overprint group is shown in (b).

specimens from site DC where movement from NRM was up to 85°.

Only one specimen from site DF was demagnetised and this only to 20 mT after which its very weak NRM became unmeasurable. For one specimen from site DG the direction continued to change (Fig. 3) and a slight increase in intensity was observed between 40 and 50

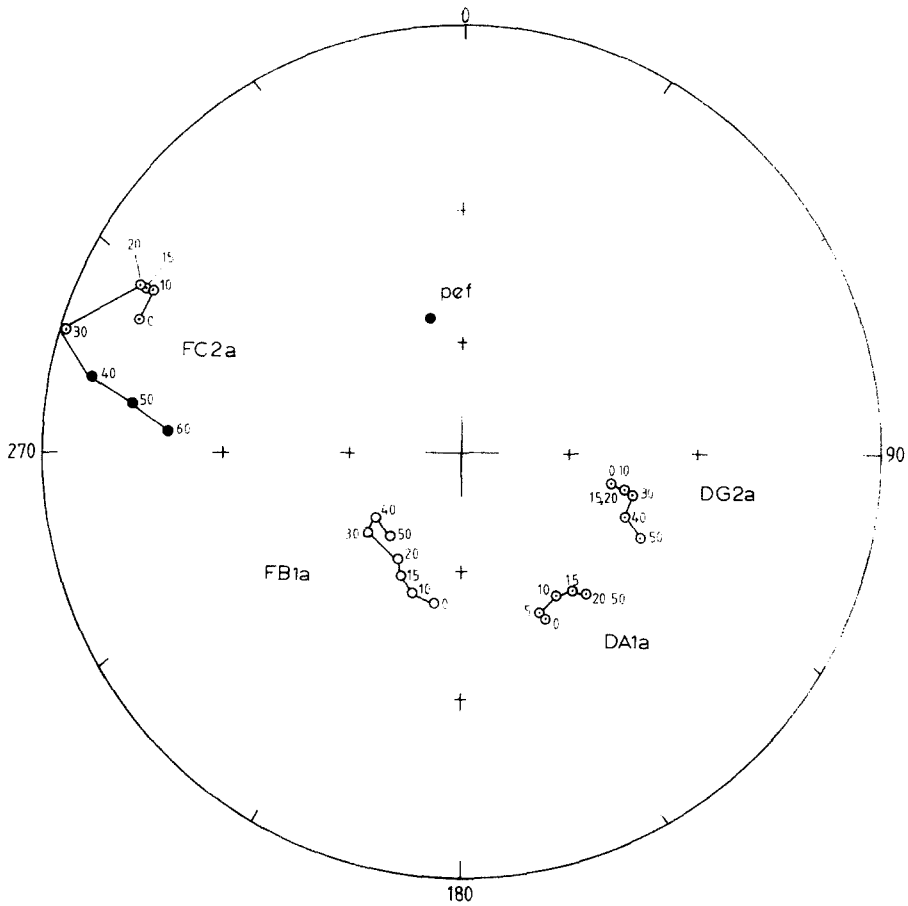


Fig. 3. Stereographic projection showing the behaviour of selected specimens during limited AFD. Symbols as in Fig. 2.

mT. No stable direction was observed for either specimen from site FB, with directional movement being approximately along great circles, in one case from $D=190°$, $I=-51°$ towards the northwest quadrant of the steronet (232°, −62°) and in the other from 250°, −34° into the northwest quadrant with a shallow positive inclination. The same type of behaviour was observed in one specimen from site FC and again a shallow positive inclination in the northwest quadrant was obtained.

For those sites for which the pilot AFD specimens showed a clear stable direction, AFD was performed on one specimen from each of the remaining samples to remove any viscous component. All AF demagnetised specimens were retained for later TD. Site mean directions after AFD are shown in Fig. 2 and summarized in Table 2. Sites FA and FD were almost unaffected by AFD but a well-defined group of eight site mean directions (sites CA, DA to DE, EA and EB) were confined to the southeast quadrant. Statistics for this group both at NRM and after AFD are given in Table 4. After AFD the group mean direction has changed by only 7° but the improvement in grouping is striking. However, after tectonic corrections (Table 1), the AFD mean direction is changed by only 7° but the grouping is substantially more dispersed. The ratio of precision parameters is 5.3 which, according to McElhinny (1964), is significant and suggests that the observed remanent direction at the eight sites after limited AFD was acquired after

TABLE 4

Mean directions and pole positions for the Chela Group

Grouping	N	Mean direction				Pole position		
		D (deg)	I (deg)	k	A_{95} (deg)	Latitude (deg)	Longitude (deg)	A_{95} (deg)
NRM AFD	8	132.0	−41.9	7.6	21.5			
AFD	8	122.9	−44.6	26.0	11.1	20.2°S	140.2°E	12.3
AFD corrected	8	115.4	−40.0	4.9	27.9			
TD group I	5	126.9	−49.3	53.3	10.6	22.1°S	146.0°E	12.2
TD group II	4	111.9	−41.1	46.9	13.6	12.5°S	132.5°E	13.7
TD group I & II (OM)	9	119.7	−45.9	40.5	8.2	17.9°S	139.8°E	9.2
TD groups I & II corrected	9	108.3	−48.0	4.5	27.2			
TD group III	3	310.9	49.3	14.5	33.6	22.8°N	31.0°W	36.7
NQ1	6	312	40	20.9	15	27°N	36°W	16

Notes: N is number of sites. A_{95} gives the circle of confidence for a pole position. NQ1 is a Nosib Group result from McWilliams and Kröner (1981) after tectonic corrections.

tilting. Application of the more valid fold test proposed by McFadden (1990) confirms this conclusion: at $N=8$ the test statistic ξ has the 95% critical value of 3.3; the *in situ* value is 1.1 while after unfolding it is 5.3.

Thermal demagnetization

The TD data from Harare were used as a guide in deciding the number and spacing of TD steps at Stanford. These varied from site to site and sometimes within sites. Stepwise thermal demagnetization was progressively reduced from initial steps of 180 to 5°C (or even less) near the Néel temperature of haematite (680°C). Components of magnetization corresponding to linear sections on orthogonal vector plots (OVPS) were determined by use of LINEFIND (Kent et al., 1983). Results at four sites are based on analyses of remagnetization circles using the method described by Jones et al. (1975) and McFadden and McElhinny (1988).

The response to TD was not uniform among the fourteen sites, but all specimens showed a dramatic decrease in magnetization close to the Néel temperature of haematite. Apart from this, behaviour during TD does suggest three groups of sites: Group I containing sites CA, DA, DB, DC, DD, FD and possibly DF; Group II with sites DE, DG, EA and EB; Group III composed of the flat-lying sites FA, FB and FC.

Group I

After removal of one or more components from specimens from Group I, OVPS showed a linear section in which the line passed through the origin. Table 3 gives site mean directions for the higher-temperature components of magnetization and the ranges over which linearity was observed is quoted for each site. The component(s) removed before linearity was achieved was (were) not consistent either within or between sites. Examples of the behaviour of specimens from Group I are shown in Fig. 4. Specimens from site DF were very weakly magnetized and TD data were of very poor quality. OVPS suggest that this site belongs to Group I and higher-temperature components were estimated manually; a site mean is given in Table 3 but this result has not been used in determining the overall mean direc-

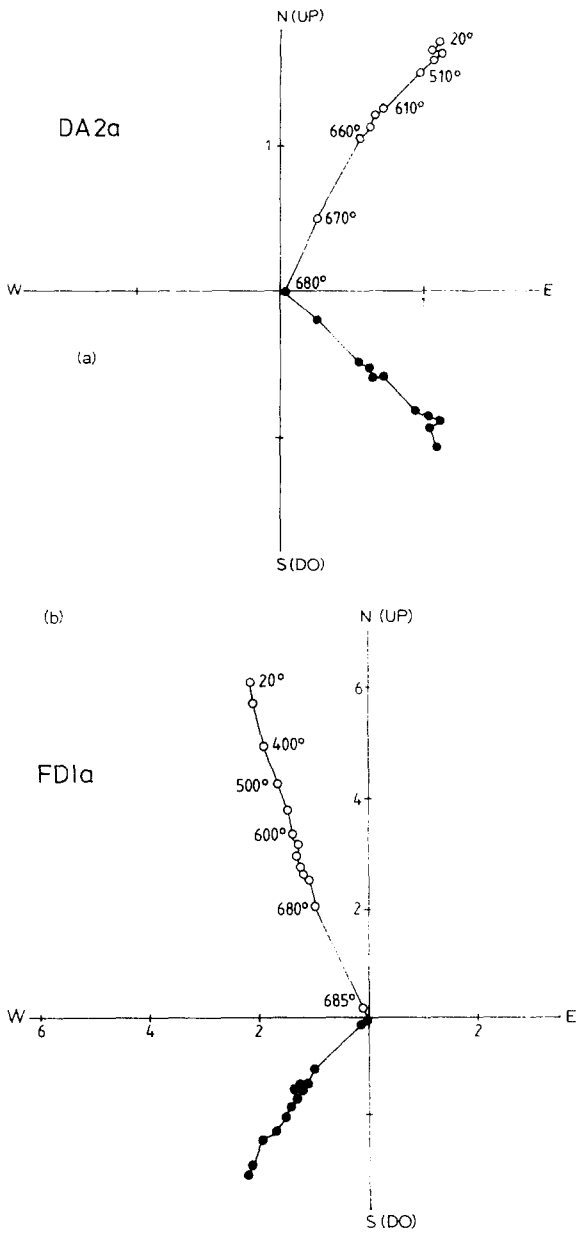


Fig. 4. Orthogonal vector plots showing TD of two specimens from Group I. Dots are plotted on the horizontal plane. Axis ticks: (a) $\times 10^{-3}$ A/m; (b) $\times 10^{-2}$ A/m.

tion. Specimen behaviour within FD was similar to that from the rest of Group I (i.e., a soft component, inconsistent in direction, overlying a single, very stable component) but the site mean direction is so different that a different (but unknown) time of magnetization is suggested.

Group II

Most specimens from Group II possessed three components of magnetization. After removal of an initial random direction, OVPS showed a linear section, often over a large temperature range. The line did not pass through the origin indicating the presence of a third component residing principally in magnetic grains having unblocking temperatures within (sometimes well within) 100°C of the Néel temperature of haematite (above the Curie temperature of magnetite). With some specimens, particularly those from site DG, the indications of a third component was evident on a stereonet where directions of magnetization lay on great circles in this highest temperature range (Fig. 5). The directions of magnetization corresponding to the linear sections on the OVPS are similar to Group I directions (except for site FD), and site mean directions, together with the ranges of linearity for this "intermediate" temperature component, are given in Table 3.

Group III

During TD no specimens from sites of Group III showed a component of magnetization similar to those isolated in Groups I and II. NRM sample directions for site FA were initially well grouped around a northerly, shallow direction that was maintained by all specimens during TD until temperature above 650°C had been attained (at which stage approximately 25% of the NRM intensity remained). Between ~650°C and ~680°C directions moved along great circles towards a northwesterly direction. All site FC specimens, in which NRM directions were widely scattered, showed evidence of two components occupying overlapping unblocking temperature ranges, and a set of demagnetization great circles was recorded (Fig. 6), suggesting a higher-temperature component with a northwesterly direction; this be-

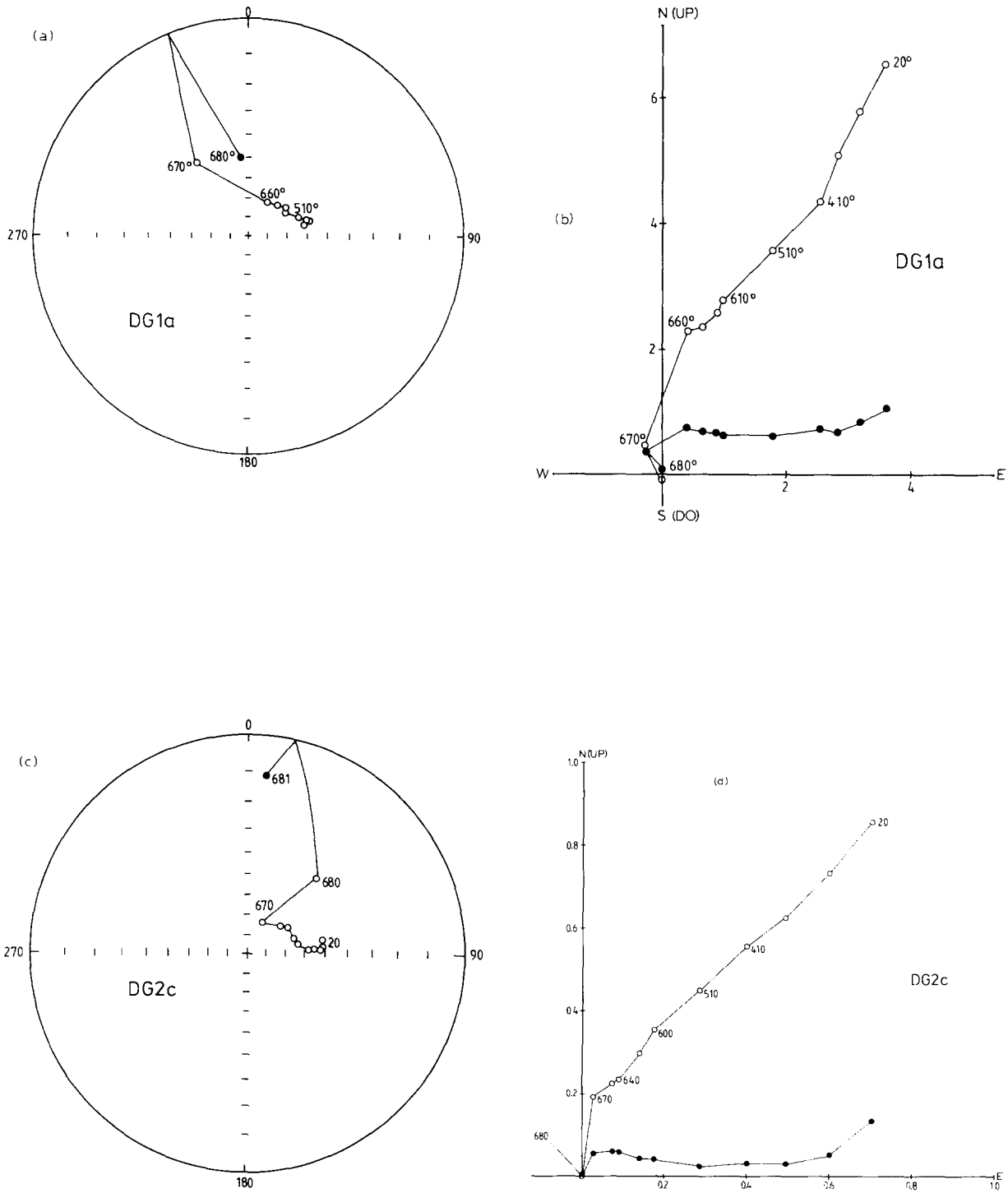


Fig. 5. Thermal demagnetization of two specimens from Group II. (a) and (c) stereographic projections; (b) and (d) OVPS. Symbols as in Figs. 2 and 4. Axis ticks: (b) $\times 10^{-3}$ A/m; (c) $\times 8.1 \times 10^{-3}$ A/m.

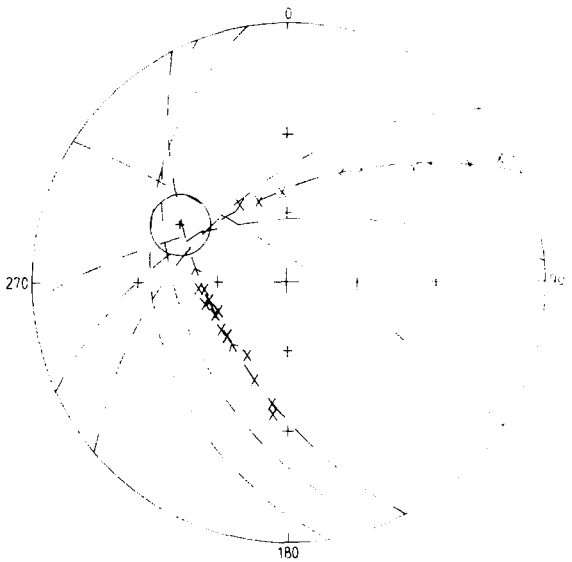


Fig. 6. Thermal demagnetization of site FC showing the remagnetization circle deduced from each specimen. Full data are shown for two specimens. The computed mean direction for the site and its circle of confidence is also shown.

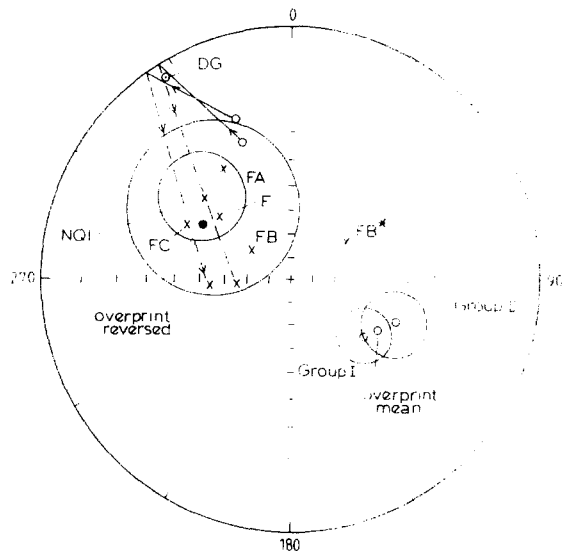


Fig. 7. Stereographic projection showing Chela Group overprint directions and the reverse of the mean overprint direction. Computed results from Group III sites are shown (see Table 5) together with their mean, F, and its circle of confidence. *NQI* is a result from the Nosib Group of Namibia (after tectonic correction). Parts of the remagnetization circles, tilt-corrected, deduced from two specimens from site DG are shown.

TABLE 5

Results from remagnetization circle analyses of Chela Group sites

Group	Site	n	D (deg)	I (deg)	k	α_{95} (deg)
III	FA	6	328.1	36.3	743	3.2
III	FB	3	304.5	68.9	497	30.0
III	FB*	8	57.1	60.0	20.0	14.7
III	FC	7	295.8	40.2	43.9	11.2
II	DG	3	327.2	− 3.7	3100	11.9

Note: n represents number of specimens used in the analysis.

haviour was anticipated by the AFD results. Specimens from site FB showed similar behaviour; however, great circles of substantial length were obtained for only three out of eight specimens. The results from each of these sites were subjected to remagnetization circle analysis and the estimate of the higher-temperature component direction is given in Table 5. We have calculated two results for site FB: FB, the result using the three great circles of substantial length, and FB*, using the data from all eight specimens. Sites FA, FB and FC each have mean directions in the northwest quadrant (Fig. 7). Great circle trajectories at high temperatures from site DG are almost parallel and short in length. Therefore, the estimate of the higher-temperature direction given in Table 5 is not as reliable as for the other three sites. After tectonic correction the trajectories from site DG pass close to both the mean of the FA, FB and FC directions and the mean of the FA, FB* and FC directions (see Fig. 7).

Discussion of results

The TD mean direction of the Group I sites is not significantly different from the mean of the TD directions recorded for Group II (see Table 4 and Fig. 7). Thus it is reasonable to combine them into a single mean direction (OM in Table 4). Tectonic corrections result in a minimal change in the mean direction but the estimate k of the precision parameter drops from 40.5 (26.0 on AFD) to 4.5 (4.9). With

$N=9$ the 95% critical value for ξ in McFadden's (1990) fold test is 3.5; after unfolding ξ is 7.6 while the in situ data give 3.6. The effect of thermal demagnetization has been to add site DG to the overprint group identified by AFD: if this site is omitted then ξ is 2.2 in situ and 4.5 unfolded (95% critical value is 3.3 with $N=8$). Clearly there is a *very* strong correlation of magnetic and tectonic directions after unfolding, indicating that the magnetization was not acquired in the unfolded position. Although the observed value of ξ is a little high with site DG included, the fold test suggests that OM magnetization was acquired post-folding.

The TD results from Group II, particularly those from site DG, suggest that these sites have not, like those of Group I, been completely remagnetized and that at the highest temperatures an underlying magnetization is present. The results from Group III, in which all three sites are flat lying, give a mean direction very close to the tilt-corrected, possibly primary, NQ1 magnetization (see Table 4) established by McWilliams and Kröner (1981) for the Nosib Group of Namibia. The tilt-corrected remagnetization circles observed at high temperatures for site DG specimens (Group II) pass close to the Group III mean and thus the Nosib result (Fig. 7).

One possible and plausible interpretation of the Chela results is that (a) the Group III mean direction (Table 4) represents the primary direction of magnetization of the Chela Group, (b) the data from the Chela Group support the correlation of the Nosib and Chela Groups, and (c) the Chela rocks have, to a greater or lesser extent, been overprinted during an event which occurred subsequent to the tilting of parts of the Chela Group.

The Chela overprint direction is approximately reversed with respect to its presumed primary direction and the NQ1. Figure 8 (after McWilliams and Kröner, 1981) shows this clearly; McWilliams and Kröner (1981) reported one site from the Nosib Group, DNF, at which only a single component of magnetization was observed and which, *before tectonic correction*, has a direction of magnetization ($D=137°$, $I=-53°$) very close to the Chela overprint. It is possible that the Chela and Nosib Groups acquired their primary magnetization contemporaneously about 1000 Ma ago [the age McWilliams and Kröner (1981) associate with NQ1], that tilting of some of the Chela strata occurred very shortly afterwards and, shortly after that, some of the Chela rocks (and site DNF in the Nosib) received their overprint following a reversal of the earth's magnetic field. This sequence appears plausible but as it conflicts with the contention of McWilliams and Kröner (1981) that the Pan-African folding of the Nosib Group (and thus DNF) began 650 Ma ago other scenarios need to be explored.

Although positive fold tests were not obtained, the data of McWilliams and Kröner (1981) support the suggestion that both components NQ1 (1000 Ma) and NQ2 (700 Ma) were acquired by Nosib rocks prior to the Pan-African folding with a later component, NQ3, less well determined, being acquired (600 Ma) post-folding. The pole position calculated for one site, FA, from its NRM mean direction is almost identical to NQ2. The Chela overprint pole falls considerably to the east of the APWP at about 600 Ma or just west of the path at about 500 Ma. A second scenario then involves correlation of the Chela and Nosib Groups with considerable overprinting of the Nosib (NQ2) and some Chela (FA) about 700 Ma ago. Tilting of both Nosib *and* Chela rocks occurred prior to about 600 Ma and subsequently (about 500 Ma) much of the Chela and some Nosib (DNF) was overprinted (again).

Interpretation of the Chela results has been made in terms of the late Proterozoic APWP presented by McWilliams and Kröner (1981). Several other late Proterozoic–early Palaeozoic APWPs have been proposed which could lead to further interpretations. The present results have been obtained from only a limited

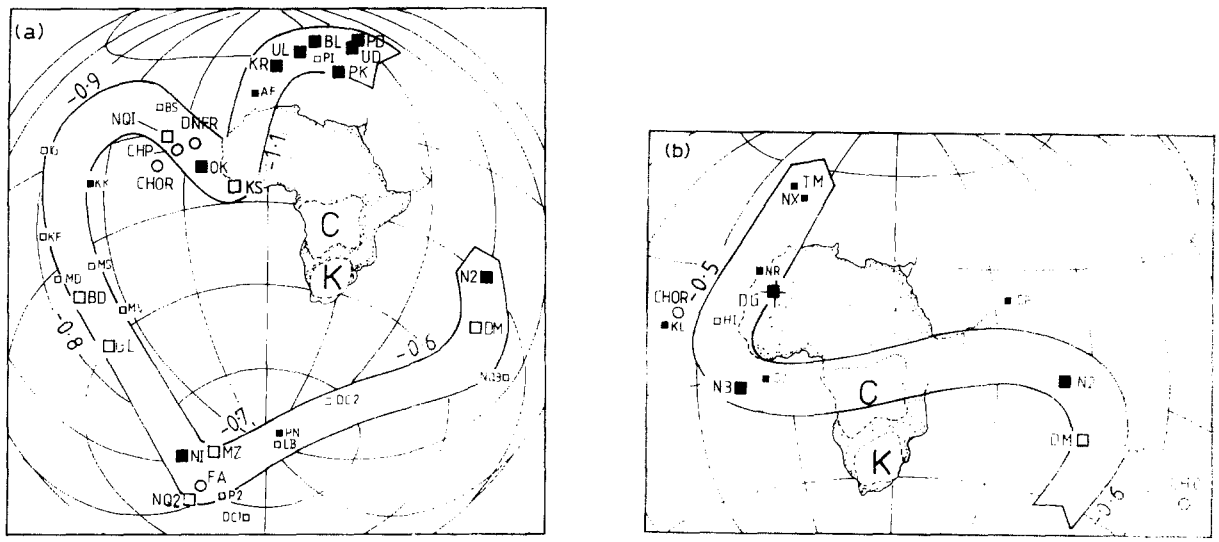


Fig. 8. Apparent polar wander paths for the Congo and Kalahari cratons from about 1100 Ma to less than 500 Ma. *CHO*=Chela overprint pole; *CHOR*=Chela overprint antipole; *CHP*=Chela primary pole position; *FA*=virtual geomagnetic pole calculated from NRM mean direction; *DNFR*=antipole of site DNF of McWilliams and Kröner (1981); *PD*=Post-Waterberg diabase, Botswana and South Africa; *UD*=Umkondo dolerites, Zimbabwe; *PK*=Premier kimberlite, South Africa; *BL*=Barby lavas, Namibia; *PI*=Plateau Series, Zambia; *UL*=Umkondo lavas, Zimbabwe; *KR*=Koras Group, South Africa; *AF*=Auborous Formation, Namibia; *KS*=Kisii Series, Kenya; *OK*=O'Okiep intrusives, South Africa; *BS*=Bukoba sandstone, Tanzania; *IG*=Ikorongo Group, Tanzania; *KK*=Klein Karas dykes, Namibia; *KF*=Kigonero flags, Tanzania; *MS*=Malagaresi sandstone, Tanzania; *MD*=Mbala dolerite, Zambia; *BD*=Bukoba dolerite, Tanzania; *MV*=Manyovu redbeds, Tanzania; *GL*=Gagwe lavas, Tanzania; *NI*=lower Nama Group, Namibia; *MZ*=Mbozi complex, Tanzania; *NQ1,2,3*=Nosib Group, Namibia; *DC1,2*=Otavi Group, Namibia; *LB*=Lake Buanji Series, Tanzania; *PN*=Pre-Nama dyke, South Africa; *DM*=Mulden Group, Namibia; *N2*=upper Nama Group, Namibia; *DP*=Doornpoort Formation, Namibia; *N3*=Nama overprint, Namibia; *SJ*=Sijarira Group, Zimbabwe; *HI*=Hook intrusives, Zambia; *KL*=Klipheuval Formation, South Africa; *DG*=Damaran granites, Namibia; *NR*=Ntonya Ring Structure, Malawi; *NX*=Blaubeker overprint, Namibia; *TM*=Table Mt. Sandstone, South Africa. For explanation of symbols and pole references see McWilliams and Kröner (1981).

number of sites which needs to be increased to provide the basis of a firm interpretation. Again, the nature of the events which caused the overprintings is not clear but low-grade, very localised and perhaps sporadic heating was probably involved. It is possible that chemical demagnetization would help cast light on the origins of the overprinting. Unfortunately, further sampling of the Chela Group (and there are some very considerable areas of Chela rocks towards the Angolan–Namibian border not yet sampled) has not been possible for many years nor is it at present.

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